

Statistics II: Introduction to Inference

Topic 2

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Point Estimator

Let us assume

- We have a population or a data-generating mechanism described by the pmf/pdf $f(x | \boldsymbol{\theta})$.
- The function f is known, but the parameter $\boldsymbol{\theta}$ is unknown.

Estimator of $\psi(\boldsymbol{\theta})$:

Let $X_i \stackrel{\text{iid}}{\sim} f(\cdot | \boldsymbol{\theta})$. Then, we call $T(\mathbf{X}) = T(X_1, \dots, X_n)$, where $T: \mathbb{R}^n \rightarrow \mathbb{R}$, is an *estimator* of the parameter of interest $\psi(\boldsymbol{\theta}) = \psi$, i.e.

$$\hat{\psi} = T(\mathbf{X}).$$

Example-1: If $f(\cdot | p) \equiv \text{Bernoulli}(p)$, $p \in (0, 1)$, then possible estimators of the parameter p include:

$$T(\mathbf{X}) = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{or} \quad T(\mathbf{X}) = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

The estimator $T(\mathbf{X})$ is called an *unbiased estimator* of $\psi(\boldsymbol{\theta})$, if

$$\mathbb{E}[T(\mathbf{X})] = \psi(\boldsymbol{\theta}), \text{ for all possible values of } \boldsymbol{\theta}.$$

Example-2: Let $X_1, \dots, X_n, n > 3$, be a random sample from a population with a true mean μ and variance σ^2 . Consider the following three estimators of $\psi(\mu, \sigma^2) = \mu$:

- $T_1 = (X_1 + X_2 + X_3)/3$,
- $T_2 = \frac{1}{8}X_1 + \frac{3}{4(n-2)}(X_2 + \dots + X_{n-1}) + \frac{1}{8}X_n$, and
- $T_3 = \bar{X}_n$

One can check that all the above estimators are unbiased estimators of μ . (Do this exercise)

Variance and Relative Efficiency (RE)

Question: When we have **two unbiased estimators**, how do we decide which one is better?

Answer: The estimator with the *lower variance* is better.

Relative Efficiency: Let T_1 and T_2 be two unbiased estimators of ψ . Then, the *relative efficiency of T_2 relative to T_1* is

$$\text{RE}(T_1, T_2) = \frac{\text{Var}(T_1)}{\text{Var}(T_2)}.$$

If $\text{RE}(T_1, T_2) < 1$, then $\text{Var}(T_1) < \text{Var}(T_2)$, that means T_1 is a better estimator than T_2 .

Recall Example-2

One can check that the variances of the estimators T_1, T_2 and T_3 are:

- $\text{Var}(T_1) = \frac{\sigma^2}{3},$
- $\text{Var}(T_2) = \frac{(n+16)}{32(n-2)}\sigma^2,$ and
- $\text{Var}(T_3) = \frac{\sigma^2}{n}$

(Do this exercise)

The relative efficiencies are:

- $RE(T_2, T_1) = \frac{\text{Var}(T_2)}{\text{Var}(T_1)} = \frac{3(n+16)}{32(n-2)} < 1$ for $n > 3$. Thus, for $n \geq 4$, T_2 is more efficient than T_1 .
- $RE(T_1, T_3) = \frac{\text{Var}(T_1)}{\text{Var}(T_3)} = \frac{n}{3} > 1$ for $n > 3$. Thus, for $n \geq 4$, T_3 is more efficient than T_1 .
- $RE(T_2, T_3) = \frac{\text{Var}(T_2)}{\text{Var}(T_3)} = \frac{n^2+16n}{32(n-2)} > 1$ for $n \geq 4$. Thus, for $n \geq 4$, T_3 is more efficient than T_2 .

Therefore, even though both T_3 and T_2 are based on all the n observations, for $n > 3$, the sample mean T_3 is more efficient than the other two estimators. (Do this exercise)

Simulation Study of Example-2 in R

```
1  # Generate n number of samples from a Unif(a, b) distribution
2  set.seed(100)
3  n = 15
4  a = 0; b = 10;
5  true.mean = (a+b)/2
6  true.var = (b-a)^2/12
7  cat("The expectation is:", true.mean, "\n")
8  cat("The variance is:", true.var, "\n")
```

```
> The expectation is: 5
> The variance is: 8.333333
```

Continued...

```
1  R = 500
2  T1 = T2 = T3 = rep(0, R)
3  for(i in 1:R){
4      x = runif(n, min = a, max = b)
5      T1[i] = mean(x[1:3])
6      T2[i] = x[1]/8 + 3/(4*(n-2)) * sum(x[2:(n-1)]) + x[n]/8
7      T3[i] = mean(x)
8  }
9  estim = cbind(T1, T2, T3)
10 mv = c(apply(estim, 2, "mean"), apply(estim, 2, "var"))
11 result = matrix(mv, nrow = 3, byrow = F)
12 colnames(result) = c("Mean", "Variance")
13 rownames(result) = c("T1", "T2", "T3")
14 cat("RE of T1 relative to T2 is:", var(T2)/var(T1), "\n")
15 cat("RE of T3 relative to T1 is:", var(T1)/var(T3), "\n")
16 cat("RE of T3 relative to T2 is:", var(T2)/var(T3), "\n")
```

```
> result
      Mean  Variance
T1 5.138960 2.8533337
T2 4.980498 0.6496339
T3 4.999723 0.5709970
> RE of T1 relative to T2 is: 0.2276754
> RE of T3 relative to T1 is: 4.997108
> RE of T3 relative to T2 is: 1.137719
```

Mean Squared Error (MSE)

Mean Squared Error (MSE): If $T(\mathbf{X})$ is an estimator of $\psi(\boldsymbol{\theta}) = \psi$, then the Mean Squared Error (MSE) is given by:

$$\text{MSE}(T(\mathbf{X})) = \underbrace{\{\mathbb{E}[T(\mathbf{X})] - \psi\}^2}_{\text{Bias}^2} + \underbrace{\text{Var}(T(\mathbf{X}))}_{\text{Variance}}$$

Question: Which one is better an **unbiased** or a **biased** estimator?

Answer: The one with the *lower MSE*.

Example-3: Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, where both the parameters are unknown. Now, consider three different estimators for $\psi(\mu, \sigma^2) = \sigma^2$:

$$T_1 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2, \quad T_2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \quad \text{and}$$

$$T_3 = \frac{1}{n+1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

(Calculate their expectations and variances)

Simulation Study of Example-3 in R

R-code to compare among the estimators T_1 , T_2 and T_3 .

```
1  set.seed(2025)
2  n = seq(10, 50, by = 1)
3  mu = 0; sigma = 1;
4  T1 = T2 = T3 = mse1 = mse2 = mse3 = c(0)
5  for(i in 1:length(n)){
6      x = rnorm(n[i], mean = mu, sd = sigma)
7      T1[i] = var(x)
8      T2[i] = ((n[i]-1)/n[i])*var(x)
9      T3[i] = ((n[i]-1)/(n[i]+1))*var(x)
10     mse1[i] = (2*sigma^4)/(n[i] - 1)
11     mse2[i] = (((n[i]-1)/n[i])^2)*mse1[i] + (sigma^4/n[i]^2)
12     mse3[i] = (((n[i]-1)/(n[i]+1))^2)*mse1[i] +
13              ((4*sigma^4)/((n[i]+1)^2))
14 }
```

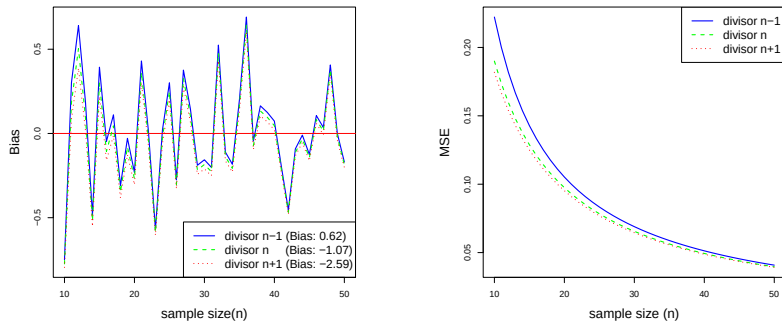


Figure 1: Plotting the biases (left) and MSEs (right) w.r.to different sample sizes for three different estimators.

Exercises

1. Do the same exercise as in **Example-2** by varying the sample size (n) and plot the REs w.r.to n in R.
2. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bin}(m, p)$. Assume that m is known. Consider two estimators of p : $\hat{p}_1 = \frac{\bar{X}_n}{m}$ and $\hat{p}_2 = \frac{S_n^2}{m} + \frac{\bar{X}_n^2}{m^2}$, where $\bar{X}_n$ = sample mean and S_n^2 = sample variance with divisor n . Show that $\hat{p}_1$ is an unbiased estimator of p , whereas $\hat{p}_2$ is a biased estimator. Also, find their relative efficiency.
3. Let $X_i \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta), i = 1, \dots, n$. We have two estimators $T_1 = 2\bar{X}_n$ and $T_2 = \frac{n+1}{n}X_{(n)}$ of θ , where $\bar{X}_n$ = sample mean and $X_{(n)} = \max\{X_1, \dots, X_n\}$. Find the expectation and variance of those estimators. Which one is better?

4. Calculate the two estimators as in **Exercise-2** using R for different sample sizes (n). (Start with $n = 5$ and take $m = 12, p = 0.38$)
 - 4.1. Plot their biases in the same graph w.r.to varying n .
 - 4.2. Plot their MSEs in the same graph w.r.to varying n .

5. Do the **Exercise-3** using R for different sample sizes (n) and plot the relative efficiency w.r.to n . (Start with $n = 10$ and take $\theta = 5$)